

SAHIL RAJESH WARKE

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PROFESSIONAL SUMMARY

Computer Engineering undergraduate and prolific open-source software architect with **60 production-grade web applications**, interactive simulation engines, and educational suites deployed on GitHub Pages. Specialist in vanilla web engineering (ES6+, DOM, Canvas, Web Audio API), Progressive Web Apps (PWA), client-side memoization, and responsive UX design. Author of a published 276-page philosophical psychology work on Internet Archive. Disciplined academic performer (**8.24 CGPA** across 4 completed semesters) actively preparing for GATE CS/IT.

TECHNICAL SKILLS

Languages:	JavaScript (ES6+), Python, C/C++, HTML5, CSS3, SQL
Frontend & UI:	React.js, Vite, HTML5 Canvas API, SVG Vector Paths, CSS Grid, Flexbox, Glassmorphic UI
Web Architecture:	Progressive Web Apps (PWA), Service Workers, Web Audio API (procedural audio), SPA Hash Routers, In-Memory Caching
Tools & Databases:	Git, GitHub Pages, MongoDB, Google Cloud Platform (GCP), LocalStorage State Engines
Specializations:	Symmetrical State Machines, Script/Phonology Engines, Data Visualization, Comparative Systems

ENGINEERING & RESEARCH EXPERIENCE

Independent Software Architect & Open-Source Creator | [Portfolio: 60 Deployed Applications](#) 2024 - Present

- Engineered, designed, and deployed **60 independent production web platforms** spanning game simulations, world linguistics, philosophy archives, creative graphics suites, and systems utilities.
- Architected offline-first Progressive Web Apps (PWAs) featuring custom Service Worker lifecycle caching, install prompts, and zero-dependency ES6+ modular structures for instantaneous load times.
- Implemented dynamic SVG vector graphing and procedural audio synthesis using the native browser Web Audio API, generating ambient music, heartbeats, and mechanical sound effects with zero external audio assets.

Author & Independent Researcher | ["The Questions That Heal" \(Published Book\)](#) Published Online

- Authored a 276-page dialogue-driven work exploring cognitive psychology, identity reconstruction, emotional healing, and epistemological inquiry through Socratic mentorship (archive.org/details/the-questions-that-heal).
- Synthesized classical Indian philosophical traditions (Advaita Vedanta of Adi Shankaracharya, Vishishtadvaita of Ramanujacharya, and Dvaita of Madhvacharya) with modern cognitive-behavioral insights.

FEATURED ENGINEERING PROJECTS

PROJECT L.I.Z. (Case File 09) Vanilla JS Web Audio API Dynamic SVG PWA [Live Application](#)

- Architected an atmospheric noir symmetrical psychological game featuring dual synchronized state machines (Detective Mode vs. Killer Mode) mutating a shared canonical crime scene model.
- Engineered an interactive forensic red-string corkboard calculating dynamic SVG Bezier curves between evidence nodes to trigger multi-clue deduction logic gates.
- Implemented real-time interrogation biometrics (simulated heart rate BPM and composure decay) paired with procedural Web Audio oscillator synthesis for dynamic tension drones.

LEADERATLAS Vanilla ES6+ SPA Hash Router Modular JSON CSS Grid [Live Application](#)

- Built a zero-dependency static web archive indexing 447+ world leaders across 199 nations, 343 historical milestones, and 15 political party genealogies.
- Developed an asynchronous client-side data loader with in-memory memoization, eliminating redundant network roundtrips across 1,000+ normalized JSON entity files.
- Created an automatic biographical cross-linking engine that dynamically scans biography strings to cross-reference treaties, factions, and ideologies into interactive entity links.

MANHWAMAKER STUDIO HTML5 Canvas Vector SVG Drag & Drop Client-Side Export [Live Application](#)

- Created a specialized digital webtoon authoring workstation supporting multi-layer canvas rendering, dynamic panel geometry re-slicing, and real-time speech balloon SVG pathing.
- Implemented Korean sound effect (Hangul Onomatopoeia) text stamping, screen tone raster shaders, and vertical-scroll canvas serialization for mobile-ready Webtoon publishing.

THE FLOW LINGUISTICS & SCRIPT SUITE (18 Platforms)

PWA

Animated SVG

Phonetics

Portfolio Archive

- Authored 18 dedicated linguistic learning suites (GreekFlow, HebrewFlow, KhmerFlow, ArmenianFlow, AmharicFlow, CyrillicFlow, StrokeFlow, etc.) decoding global alphabets and abugidas.
- Integrated stroke-order vector animations, IPA phonetic guides, Devanagari/Ge'ez script rules, spaced-repetition flashcards, and full offline Service Worker caching.

COREZEN - Holistic Fitness & Movement Architecture

PWA

Web Audio API

LocalStorage

Live Application

- Architected a privacy-first, offline-ready fitness operating system featuring 4 specialized training modalities: Calisthenics Progression, Yoga Asana & Pranayama, Powerbuilding, and Postural Rehab.
- Engineered a custom workout logging engine with automated volume load computation, RPE telemetry, rest interval chimes via Web Audio API, and zero-cloud local data persistence.

UNITYSCRIPT

Python

Gemini API

Google Cloud

JavaScript

AI Exploration Platform

- Developed an AI-powered comparative literature and sacred scripture exploration platform integrating Large Language Models (Gemini API) for contextual multi-tradition cross-referencing.
- Implemented semantic text retrieval pipelines to parse philosophical queries and generate contextual insights across historical commentaries.

EDUCATION

Bachelor of Computer Engineering (B.E.) MET's Institute of Engineering, Bhujbal Knowledge City		2024 - 2028 (Expected)
Cumulative CGPA: 8.24 / 10.00 (Sem 1: 8.72 Sem 2: 7.83 Sem 3: 8.05 Sem 4: 8.36 Sem 5: Savitribai Phule Pune University)		
Higher Secondary Certificate (HSC) Class XII - Maharashtra State Board		73.50%
Secondary School Certificate (SSC) Class X - Maharashtra State Board		87.00%

ACADEMIC COURSEWORK & GATE CS/IT FOUNDATION

Core Systems:	Operating Systems, Computer Organization & Architecture, Database Management Systems (DBMS)
Theory & Algorithms:	Data Structures & Algorithms, Theory of Computation (Automata & Formal Languages), Discrete Mathematics
Networks & Security:	Computer Networks (TCP/IP, OSI, Socket Programming), Information & Network Security
Competitive Prep:	Actively preparing for GATE CS/IT; comprehensive problem-solving in discrete math, algorithms, and system design